
Cost-Optimized Menu Recommendation System using Machine Learning

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The Problem

Currently, consumers lack tools to forecast these price changes, which often leads to “market shock,” where the prices of raw ingredients in the market are higher than expected. This results in inefficient budget planning because there is no system that connects external factors with practical food menu recommendations.

Research Question

- 1.How can the system analyze food prices and external factors to forecast prices and recommend cost-efficient meals?
- 2.Which factors (seasonality, fuel, festivals) most affect price changes for each ingredient?
- 3.What is the optimal forecasting horizon (e.g., 1 week vs. 1 month) while maintaining accuracy?
- 4.How much can such a system reduce food expenses compared to traditional planning?



Objectives of the Study

To develop a machine learning-based price forecasting and cost-optimized menu recommendation system designed to mitigate the impact of fresh food price volatility on consumers and small business operators.



External factors impact on food price volatility



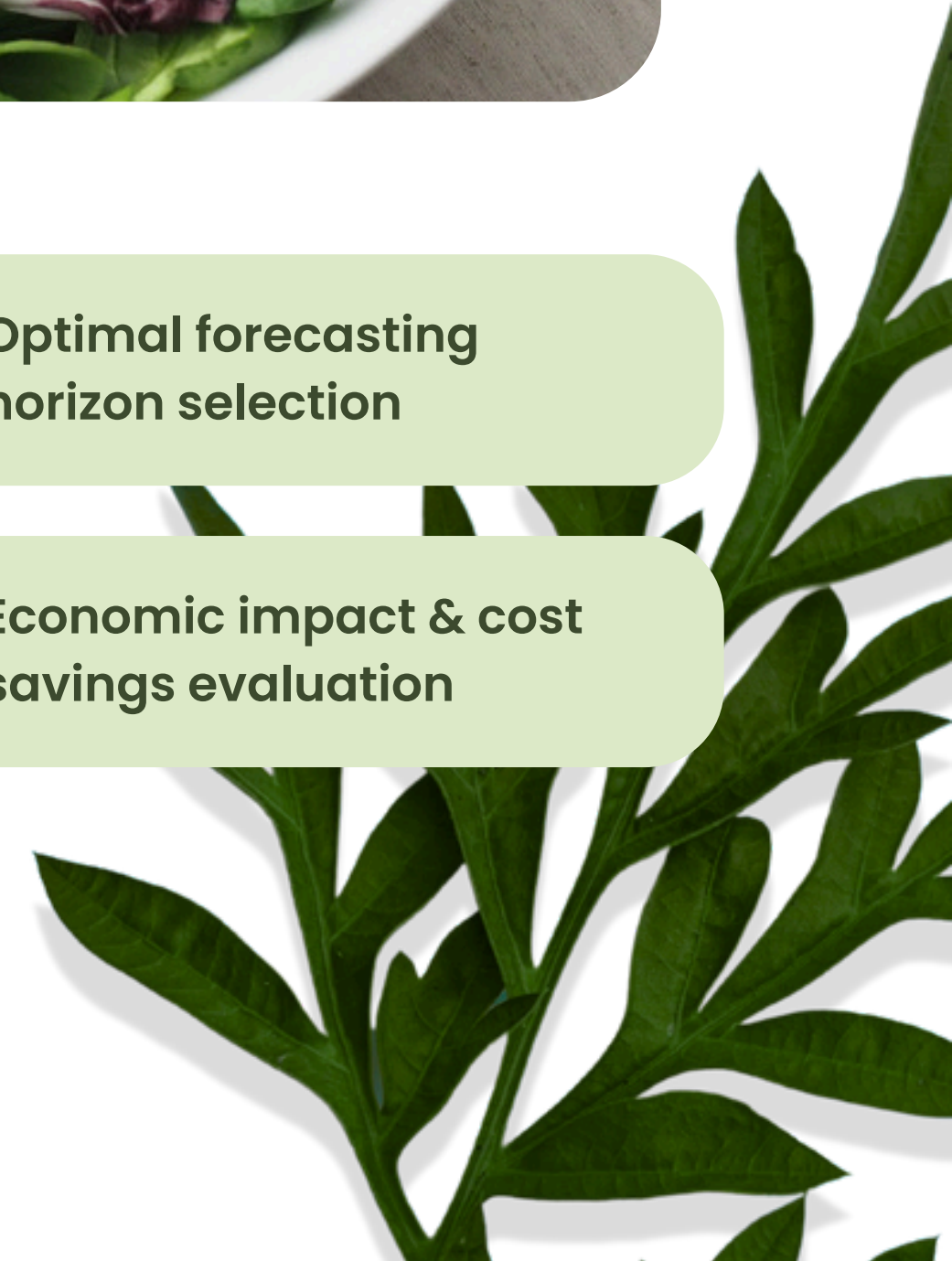
Optimal forecasting horizon selection



Cost-optimized meal recommendation algorithm



Economic impact & cost savings evaluation



LITERATURE REVIEW

Paper / Author	Focus / Methodology	Key Findings & Results
Chen (2021) & Zhao (2024)	Shifted from traditional ARIMA to Machine Learning for agricultural prices.	ML handles market volatility and non-linear data better than statistical models.
Le Ngoc (2023) & Watson (2024)	Evaluated Random Forest and Deep Learning models on multivariate data (weather, fuel).	Tree-based models process exogenous variables effectively, while Deep Learning captures long-term trends.
Our Research Gap	Previous works focus purely on theoretical point forecasting.	Contribution: Transform forecasts into a practical Cost-Optimized Menu System and handle structural missing data.

Scope and Limitations

🔍 Project Scope:

Timeframe: 5-year historical dataset (Jan 2021 – Dec 2025), totaling 1,827 days, allowing the model to fully capture long-term market trends and annual seasonality.

Spatial Scope: Weather data from 4 strategic provinces (Pathum Thani, Ratchaburi, Phetchabun, Nakhon Pathom).

Target Variables: 13 key fresh ingredients (proteins, vegetables, herbs).

Forecasting Horizon: 7 days in advance.

Output Scope: Cost ranking for 5 popular Thai prototype menus.



🔍 Limitations:

Weekend Data Gaps: Market prices are only published on weekdays. .

Static Recipe Proportions: Cost calculations use standard, fixed ingredient amounts and cannot adjust for personal preferences or diet variations.

Limited Economic Scope: The model focuses on weather and diesel costs, excluding broader factors like national inflation or political events.

Incomplete Ingredient Coverage: The model predicts prices for only 13 key ingredients. Complex dishes requiring many unlisted ingredients may not reflect their true total cost, which could slightly affect the accuracy of the menu ranking.



Methodology

Data Pipeline & Preprocessing

Multivariate Time-Series: A 5-year daily dataset (1,827 records) integrating four distinct data sources:

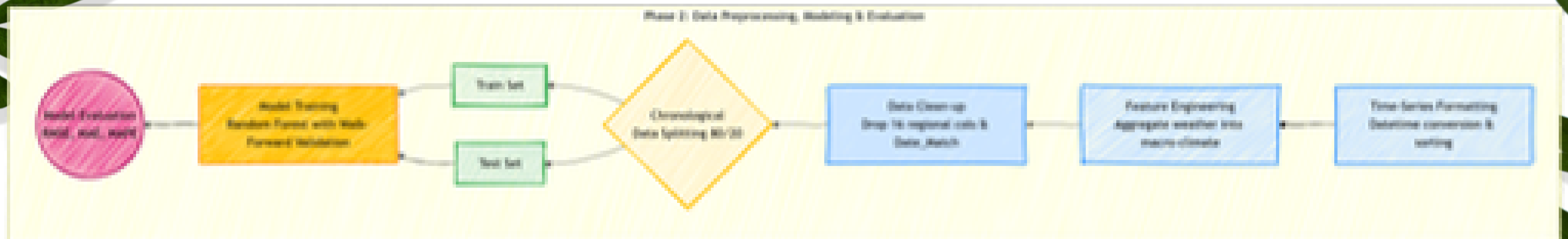
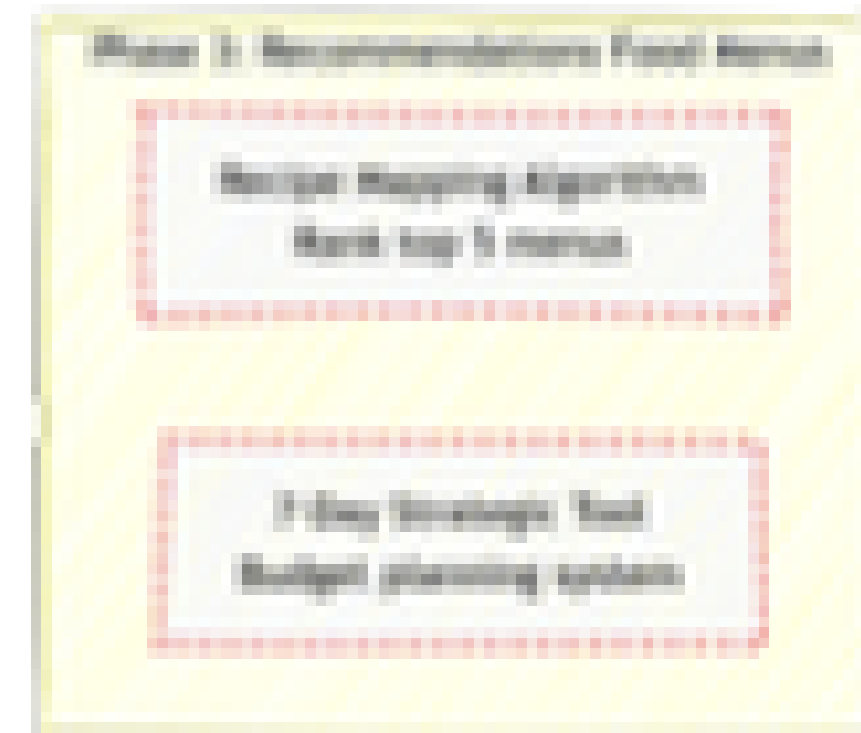
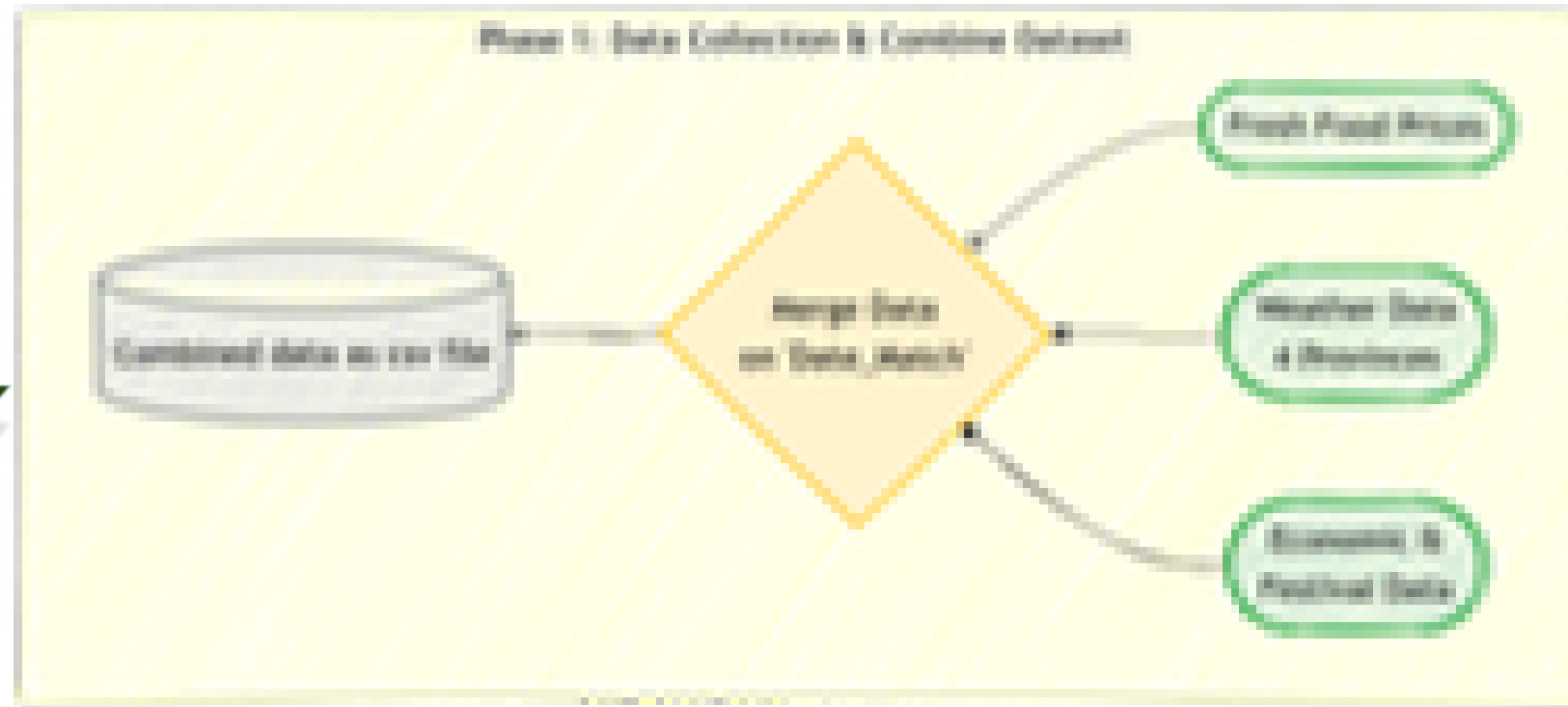
- **Target Prices:** 13 fresh ingredients from the Department of Internal Trade (DIT).
- **Spatial Weather:** Temperature, rainfall, humidity, and soil moisture from Open-Meteo across 4 key agricultural provinces.
- **Logistics Anchor:** Daily high-speed diesel prices from Bangchak Corporation.
- **Contextual Calendar:** Custom tracking of Thai public holidays and significant cultural festivals.

Handling Structural Missingness: Agricultural market prices are strictly published on official business days, resulting in systematic missing values during weekends and public holidays.

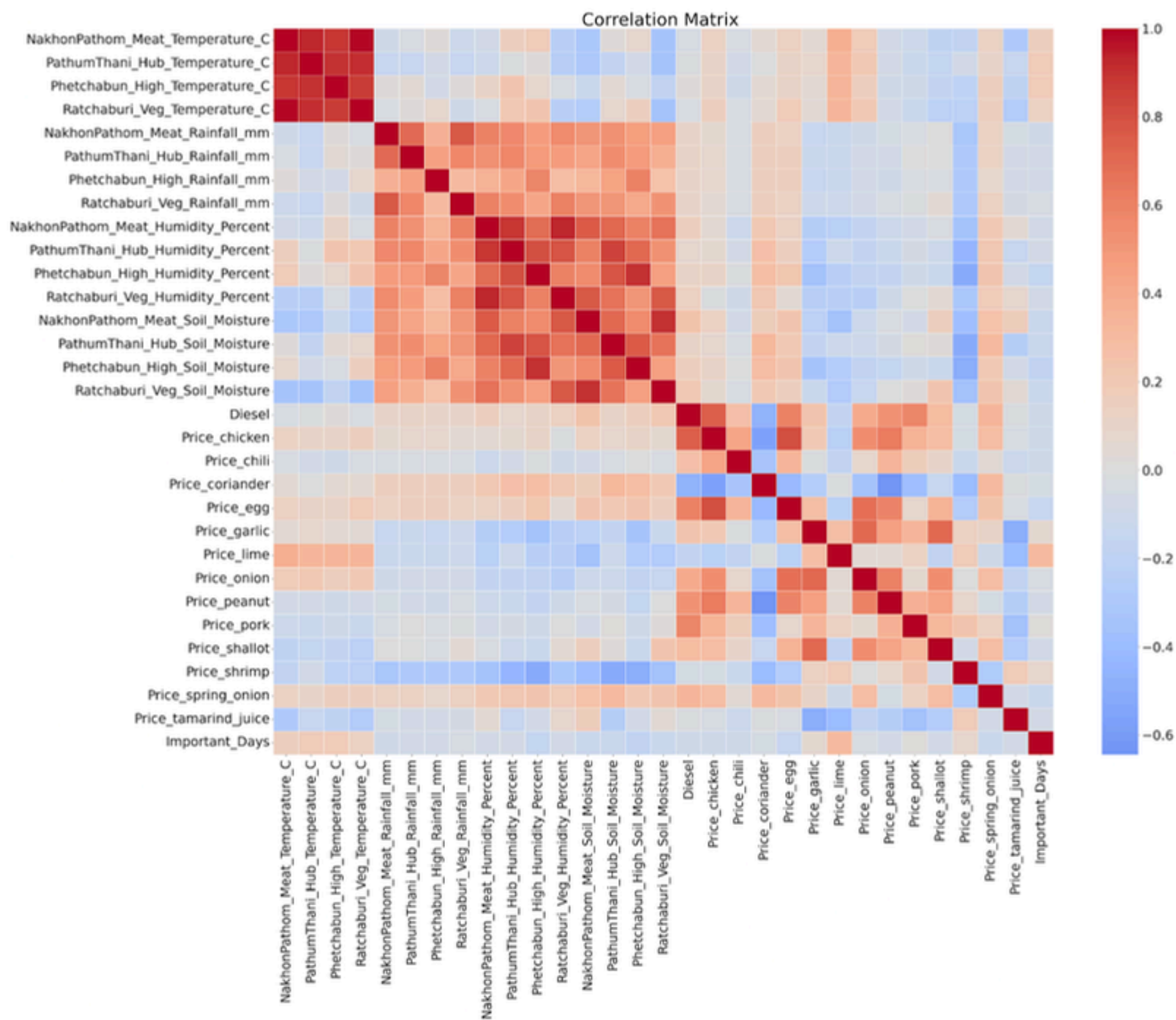
- **The Solution (Forward Fill – ffill):** Rather than using standard interpolation which causes data leakage (peeking into future prices), the system applies a Forward Fill technique. We assume weekend market prices remain stagnant at Friday's closing price, perfectly reflecting real-world consumer purchasing realities while strictly preserving chronological continuity.



Architecture

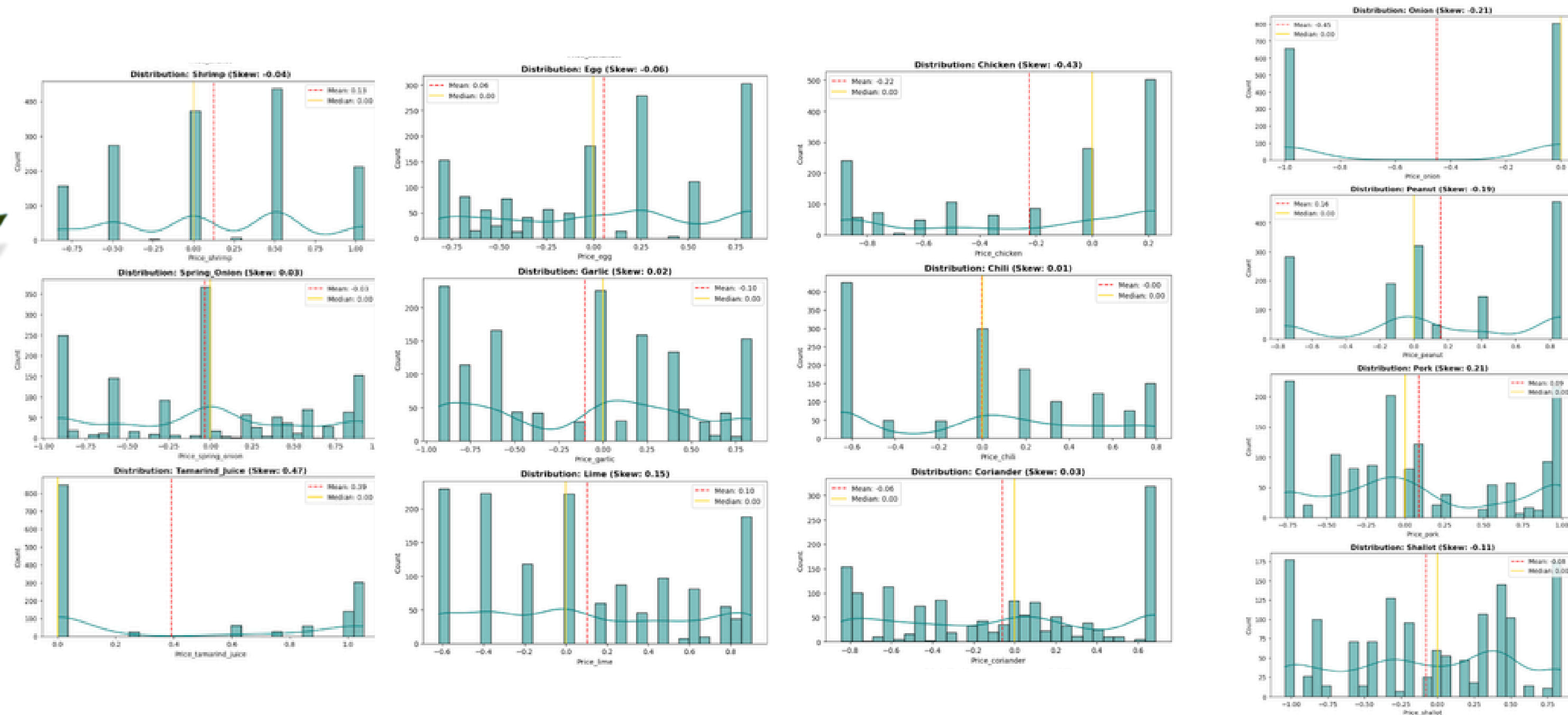


Pearson Correlation Heatmap of All Features



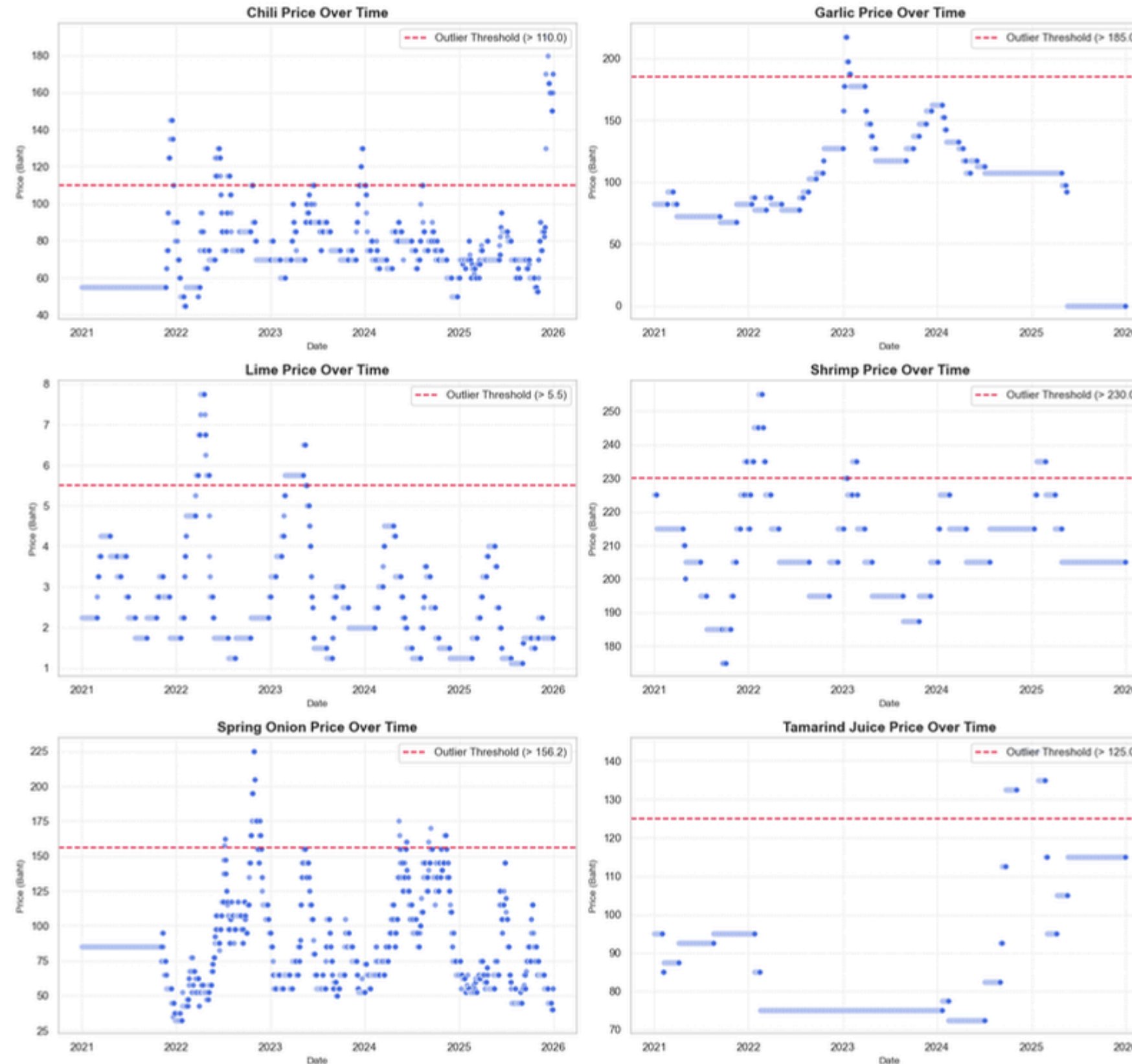
The correlation matrix highlighting severe spatial multicollinearity among provincial weather variables (dark red clusters, $r > 0.8$) and the macroeconomic influence of diesel prices on livestock.

Distribution and Skewness of Target Variables



The distribution plot illustrates that, after feature transformation and capping, the data has been standardized and is now more symmetrical. This helps reduce skewness and minimizes potential bias in the model, leading to more stable and reliable predictions.

Time-Series Volatility and Outlier Thresholds



Longitudinal time-series plots with predefined outlier thresholds (red dashed lines). The visuals confirm that price spikes are episodic and non-linear, often aligning with festivals or seasonal droughts.



Model Evaluation & Validation

Architectures Tested

Systematically evaluated the baseline Random Forest Regressor against deep sequential learning models (SimpleRNN and LSTM) to determine the optimal forecasting engine.

The Optimal Model

Model	MAE(THB/kg)	RMSE(THB/kg)	MAPE (%)
Random Forest	19.65	28.58	17.27%
SimpleRNN	25.24	37.42	18.75%
LSTM	21.42	31.52	21.85%

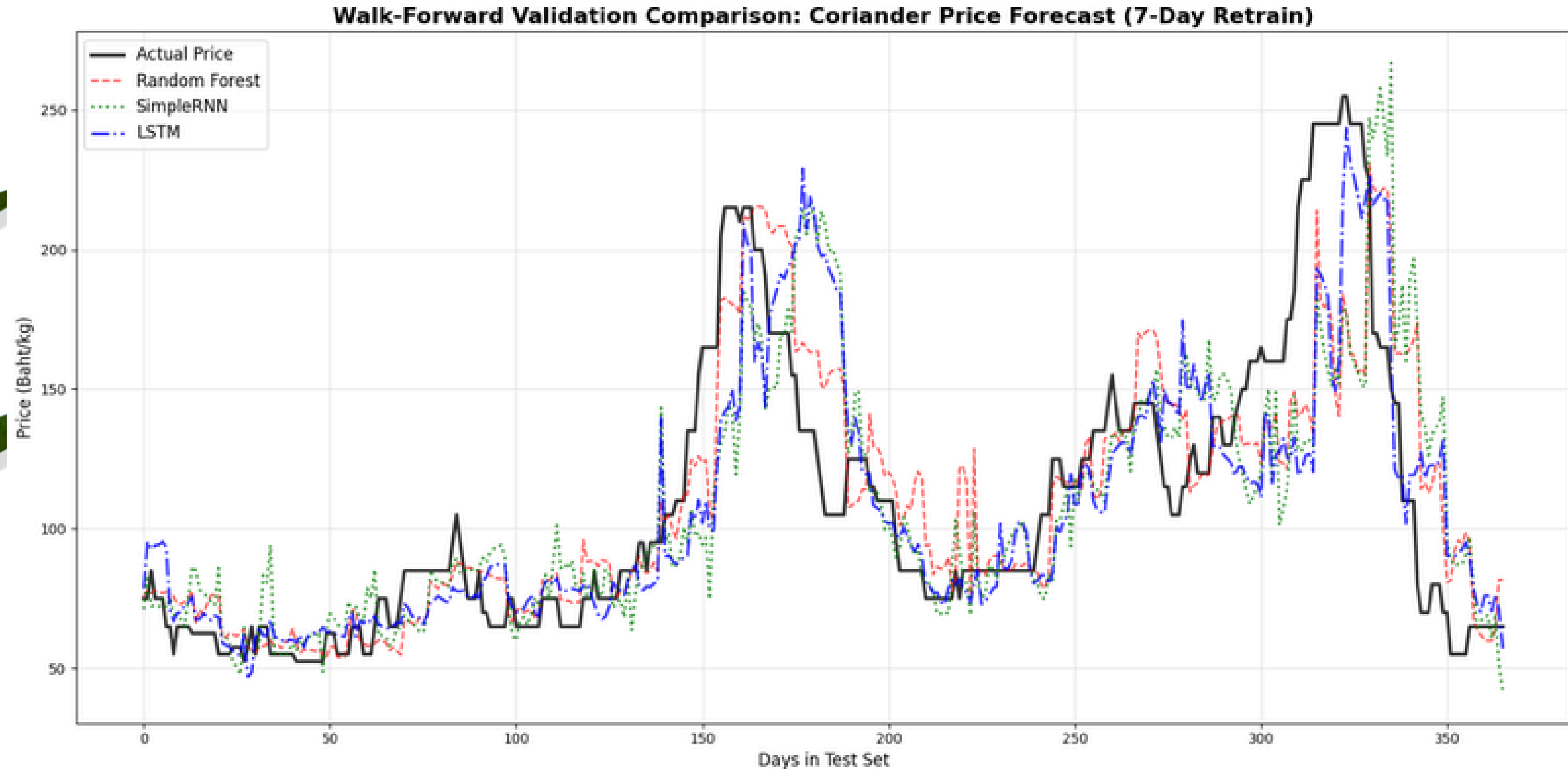
Contrary to the initial hypothesis that deep learning would dominate, **the Random Forest Regressor** significantly outperformed the neural networks across all error metrics.



Best Parameters and Feature Importance

Model	Best Parameters	Top 3 Input Features
Random Forest	Best Parameters: {'max_depth': 20, 'min_samples_leaf': 1, 'min_samples_split': 2, 'n_estimators': 100}	Price_chicken, Price_spring_onion, Price_peanut
SimpleRNN	units: 64, dropout: 0.2, epochs: 50	Price_shallot, Price_onion, Price_tamarind_juice
LSTM	units: 64, dropout: 0.2, epochs: 50	Price_tamarind_juice, Avg_Temperature_C, Avg_Soil_Moisture

Walk-Forward Validation Comparison (Actual vs. Predicted)



The superior performance of **the Random Forest** stems from its tree-based structure. It successfully learned the non-linear thresholds of discrete tabular features (**like the Important_Days flag**) and robustly tracked extreme price spikes. In contrast, SimpleRNN exhibited high variance (overshooting), while LSTM overly smoothed the predictions, failing to capture the true magnitude of market shocks.



Discussion

Interpreting Feature Importance



- **Tree-based & RNN Models:** Heavily rely on contemporaneous ingredient prices (e.g., Chicken, Shallot). This implies market shocks are closely tied to shared supply chain dynamics.
- **Deep Sequential Models (LSTM):** Prioritize continuous environmental variables (Temperature, Soil Moisture). This shows LSTM excels at capturing long-term climatic trends over sudden market spikes.

Why Random Forest Wins & Feature Insights



Model Performance

- **Random Forest (Optimal):** The tree-based architecture excels at learning non-linear thresholds, allowing it to effectively capture sudden "Market Shocks" (extreme price spikes).
- **LSTM / SimpleRNN:** Deep learning models tend to "smooth out" predictions, which frequently results in underestimating the magnitude of extreme price volatility.



Conclusion

Conclusion & Key Takeaways



- 1. Robust Data Pipeline:** Successfully resolved structural missingness (weekend/holiday data gaps) using the Forward Fill technique, preserving chronological integrity without causing data leakage.
- 2. The Optimal Engine:** Random Forest proved to be the most effective model for forecasting highly volatile agricultural prices, outperforming complex deep learning models (RNN/LSTM).
- 3. Actionable Application (Main Contribution):** Successfully translated theoretical price forecasts into a practical "Cost-Optimized Menu Recommendation" system. This empowers users to proactively avoid overpriced ingredients and efficiently manage their weekly food budgets.

Future Work & System Expansion



- 1. Addressing Incomplete Ingredient Coverage:** Currently, the model predicts prices for 13 key ingredients. As a result, the calculated total cost for highly complex dishes requiring numerous untracked ingredients may be underestimated. Future iterations must expand the data collection pipeline and feature space to track a much wider array of minor vegetables, herbs, and condiments, ensuring absolute precision in recipe cost ranking.
- 2. User Personalization:** Integrate customized user preferences—such as dietary restrictions, allergy alerts, and flavor profiles—by bridging the quantitative price model with qualitative consumer survey data.
- 3. Full-Scale Deployment:** Transition the current static mock-up into a fully deployed, live web or mobile application, enabling real-time API data synchronization and seamless weekly meal planning for end-users.

Our Deployment

7-Day Predicted Price

Ingredient price forecasting and budget planning system for the next 7 days

#	Ingredient	Current	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7	Trend
1	chicken	65.00	65.00	65.00	65.50	65.50	66.00	66.00	66.00	— Stable (+15%)
2	chili	140.00	142.00	145.00	148.00	150.00	155.00	155.00	160.00	▲ Price Up (+14.3%)
3	coriander	100.00	105.00	108.00	110.00	115.00	115.00	118.00	120.00	▲ Price Up (+20.0%)
4	egg	4.10	4.10	4.10	4.10	4.15	4.15	4.15	4.15	— Stable (+12%)
5	garlic	90.00	89.00	88.00	86.50	85.00	85.00	84.00	83.00	▼ Price Down (~7.8%)
6	lime	1.50	1.55	1.60	1.65	1.70	1.75	1.80	1.85	▲ Price Up (+23.3%)
7	onion	35.00	34.50	34.00	33.50	33.00	32.50	32.00	32.00	▼ Price Down (~8.6%)
8	peanut	77.00	77.00	77.00	77.50	77.50	77.50	78.00	78.00	— Stable (+13%)
9	pork	160.00	162.00	164.00	165.00	166.50	167.50	168.00	170.00	▲ Price Up (+6.3%)
10	shallot	68.00	67.50	67.00	66.50	66.00	65.50	65.00	64.00	▼ Price Down (~5.9%)
11	shrimp	205.00	205.00	205.00	205.00	206.00	206.00	205.00	205.00	— Stable (0.0%)
12	spring_onion	40.00	41.00	42.00	43.00	44.00	45.00	46.00	47.00	▲ Price Up (+17.5%)
13	tamarind_juice	115.00	115.00	115.00	115.00	115.00	115.00	115.00	115.00	— Stable (0.0%)

* Predicted price data is automatically updated from the JSON file.

[Go to Menus](#)

Menu Value Prediction

Ranking menus based on the highest expected cost reduction (Percentage Change)

Select Forecast Day:

Day 1 (Monday)

[Back to Dashboard](#)



#1 Best Value

Tom Yum Goong (Spicy Shrimp Soup)

Ingredients: Fresh shrimp, lime, chili

฿86.02

+0.14%

Cost Breakdown:
Fresh shrimp (300g): ฿61.50 (from ฿61.50)
Lime (2 pcs): ฿3.10 (from ฿3.00)
Chili (10g): ฿1.42 (from ฿1.40)



#2

Kai Jeow Moo Sap (Pork Omelette)

Ingredients: 3 eggs, ground pork, green onion

฿40.81

+0.27%

Cost Breakdown:
Eggs (3 pcs): ฿12.30 (from ฿12.30)
Ground pork (50g): ฿8.10 (from ฿8.00)
Green onion (10g): ฿0.41 (from ฿0.40)



#3

Khao Pad Moo (Pork Fried Rice)

Ingredients: Pork, eggs, garlic

฿53.84

+0.53%

Cost Breakdown:
Pork (150g): ฿24.30 (from ฿24.00)
Eggs (2 pcs): ฿8.20 (from ฿8.20)
Garlic (15g): ฿1.33 (from ฿1.35)



#4

Pad Kra Pao Moo Sap (Basil Pork)

Ingredients: Ground pork, garlic, chili, eggs

฿79.56

+0.77%

Cost Breakdown:
Eggs (2 pcs): ฿8.20 (from ฿8.20)
Pork (300g): ฿48.60 (from ฿48.00)
Garlic (15g): ฿1.33 (from ฿1.35)
Chili (10g): ฿1.42 (from ฿1.40)



#5

Larb Moo (Pork Salad)

Ingredients: Ground pork, shallots, lime, green onion

฿73.77

+0.93%

Cost Breakdown:
Ground pork (300g): ฿48.60 (from ฿48.00)
Shallots (30g): ฿2.02 (from ฿2.04)
Lime (15 pcs): ฿2.33 (from ฿2.25)
Green onion (20g): ฿0.82 (from ฿0.80)

<http://192.41.170.112:5843/index.html>

**Thank You
Very Much!**

